

# Predicting the Trajectory of the UK Gender Pay Gap Using Machine Learning

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## Abstract

Nine consecutive years (2017–2025) of mandatory UK gender pay gap (GPG) disclosure data are analysed, covering approximately 89,000 firm-year observations from roughly 15,000 unique employers. The primary task is a three-class trajectory classification: predicting whether a firm’s median hourly pay gap will improve, stagnate, or worsen relative to the following year, using only a single-year structural snapshot as input. Six classifiers are evaluated, from a stratified dummy baseline through Gradient Boosting. The best model achieves a holdout macro-F1 of 0.502 on 2023–2024 test data, against a dummy floor of 0.329. Tree ensembles outperform linear models because the decision boundary separating trajectory classes is non-linear in the quartile, bonus, and sector feature space. The modest absolute performance is the central finding: mandatory disclosure has produced only marginal, unsystematic year-to-year movement, leaving little forecastable signal for any model to recover. Vertical segregation (the female share of the top pay quartile) and bonus culture emerge as the dominant structural predictors of trajectory, consistent with established economic explanations of the gender pay gap.

## 1 Introduction

### 1.1 Contextual Background

Despite decades of equal pay legislation, women continue to earn less than men in virtually every economy in the world. This persistent disparity, the gender pay gap (GPG), is not merely a statistical artefact: it reflects deep-rooted asymmetries in labour market structure, occupational segregation, and organisational hierarchies that resist easy policy fixes. Understanding *why* the gap persists, *where* it is largest, and *which firms* are closing it requires moving beyond aggregate national statistics. Machine learning offers a powerful toolkit for this purpose.

The GPG is the most widely cited indicator of labour-market gender inequality. In its unadjusted form, it expresses the difference between the average earnings of men and women as a percentage of men’s earnings, without controlling for individual or job characteristics [6]. Two complementary measures are typically reported. The **mean hourly gap** captures the difference between average male and female hourly earnings and is sensitive to high-income outliers, making it informative about disparities at the top of the distribution. The **median hourly**

**gap** compares the typical male and female worker and is more robust to outliers; it is the primary headline indicator used by the UK Office for National Statistics [11]. The two measures often diverge meaningfully: a firm with a small median gap can still record a large mean gap if a handful of senior men earn substantially more than senior women, a pattern central to the “glass ceiling” literature.

The United Kingdom occupies a unique position in this debate. Since 6 April 2017, the *Equality Act 2010 (Gender Pay Gap Information) Regulations 2017* [17] have required every employer in Great Britain with 250 or more employees to publish six standardised GPG indicators each year on a public government portal: the mean and median hourly pay gaps, the mean and median bonus pay gaps, the proportion of men and women receiving bonuses, and the proportion of men and women in each pay quartile. Non-compliance can trigger enforcement action by the Equality and Human Rights Commission and, ultimately, unlimited fines. The result is one of the largest publicly available firm-level pay-gap datasets in the world, covering roughly 10,000 employers per year and now spanning nine reporting cycles (2017–2025).

Recent UK figures show slow but visible convergence. According to the ONS [11], the median GPG for full-time employees stood at **6.9%** in April 2025, down from 7.1% in 2024 and from roughly 17.4% in 1997. Across all employees (full- and part-time combined) the gap was 12.8% in 2025, compared with 13.1% the previous year. At the European level, Eurostat [6] reports an EU-27 unadjusted gender pay gap of **11.1%** in 2024, with substantial heterogeneity across Member States, from  $-0.8\%$  in Luxembourg to 18.8% in Estonia.

This UK transparency framework is no longer an isolated experiment. Directive (EU) 2023/970 on pay transparency, adopted in June 2023 [16], will require all EU Member States to transpose comparable disclosure obligations into national law by 7 June 2026. The UK regime therefore offers an early-mover, nine-year laboratory for studying how mandatory transparency interacts with firm structure and gender pay outcomes.

## 1.2 Objective of the Study

The central predictive task is to determine whether a firm’s median hourly pay gap will **improve**, **stagnate**, or **worsen** relative to the following year, using only a single-year structural snapshot as input. The classification framing is preferred over regression because it produces actionable, policy-legible labels: regulators can identify firms likely to worsen and target interventions accordingly. The longitudinal depth of the panel is its central asset: it allows us to model not only the *level* of the gap as a function of firm characteristics, but also its *direction of travel* over time, an angle under-exploited in the existing literature.

## 1.3 Research Questions

The empirical analysis is organised around three formal research questions spanning unsupervised and supervised paradigms.

**RQ1 (Unsupervised learning):** What latent structural archetypes of firms emerge from pay-quartile composition and bonus participation data alone, and how do these profiles relate to observed gender pay gap levels? This question is addressed via PCA followed by  $K$ -means clustering on ten standardised numeric features.

**RQ2 (Supervised classification):** Can a

single-year structural snapshot predict whether a firm’s median hourly pay gap will **improve**, **stagnate**, or **worsen** the following year? The target variable `trajectory_label` is constructed from the per-employer next-year change in `DiffMedianHourlyPercent`: improving ( $\leq -1.5$  pp), worsening ( $\geq +1.5$  pp), stagnant otherwise. Trajectory prediction is expected to be harder than level prediction; sectoral baseline and initial quartile imbalance should nevertheless carry signal.

**RQ3 (Structural drivers of trajectory):** Which firm characteristics most strongly predict the *direction* of next-year change in the median hourly pay gap? Answered through feature-importance analysis (permutation importance, logistic-regression coefficients, and SHAP values) on the RQ2 classifier.

Together, RQ1 maps **how** firms cluster into recognisable structural profiles, RQ2 asks **whether** and **for whom** the gap is changing, and RQ3 identifies **what** structural characteristics drive its direction of travel.

## 2 Literature Review

### 2.1 Machine Learning in Labour Economics

The methodological case for using ML in this domain is set out by Mullainathan and Spiess [9] in the *Journal of Economic Perspectives*. They distinguish *prediction problems* (the natural province of ML) from *parameter estimation* (the traditional concern of econometrics) and argue that many policy-relevant labour-market questions — including identifying which firms are most likely to deviate from a benchmark — are fundamentally predictive. Their framework legitimises the use of regularised and tree-based models, while flagging well-known caveats around interpretability and inferential validity.

Building on this methodological turn, Strittmatter and Wunsch [12], in *Labour Economics*, revisit the estimation of the German gender pay gap using a post-double-LASSO procedure. They show that ML-augmented covariate selection — including high-order interactions and second-order polynomials that an analyst would rarely specify by hand — meaningfully changes the estimated adjusted gap, especially at the tails of the wage distribution. Their result is a strong empirical argument that ML is not merely a predictive convenience but can alter substantive economic conclusions.

Caprettini, Manieri, and Romeo [15] provide a direct application to the gender pay gap, using gradient-boosted trees on Italian firm-level pay data and showing that ML-derived feature importance yields a richer gap decomposition than traditional Oaxaca–Blinder methods. The collective lesson is that ML methods consistently improve predictive accuracy on labour-market outcomes, can change the substantive interpretation of the gap, and are most informative when paired with careful feature engineering.

## 2.2 Gender Pay Gap: Drivers and Evidence

The modern economic literature on the GPG is anchored by Blau and Kahn [2], whose comprehensive review in the *Journal of Economic Literature* documents the long-run convergence of male and female earnings in the United States between 1980 and 2010 and decomposes its remaining residual. Their central finding is that conventional human-capital variables (education, experience) explained very little of the unadjusted gap by 2010, while **occupation and industry** had become the dominant observable contributors. Crucially, the gap narrowed least at the top of the wage distribution, foreshadowing the glass-ceiling literature. This decomposition logic motivates RQ1’s emphasis on sectoral and structural features.

Goldin [7], in her presidential AEA address, refines the explanation by attributing much of the residual gap to the **non-linear pricing of working hours** in occupations that demand temporal inflexibility, typically high-skilled, client-facing roles in finance, law, and consulting. Her framework predicts that gaps will be largest in firms and sectors that reward long, continuous, always-available careers, and smallest where work can be substituted across colleagues, a pattern directly visible in the sectoral heterogeneity of the UK disclosure data.

Arulampalam, Booth, and Bryan [1] extend the analysis across the wage distribution. Using harmonised EU Household Panel data for ten countries, they apply quantile regression and demonstrate that gender pay gaps are systematically larger at the top of the distribution (“glass ceilings”) and, in some countries, also at the bottom (“sticky floors”). Glass ceilings dominate. This distributional perspective directly justifies the project’s reliance on quartile-composition features (`FemaleTopQuartile` etc.) as inputs to both supervised and unsupervised models.

Card, Cardoso, and Kline [4], in the *Quarterly*

*Journal of Economics*, shift the level of analysis from individuals to **firms**. Using matched employer–employee Portuguese data, they show that firm-specific pay premiums are an important source of wage inequality and that women are systematically less likely to work at high-premium firms (sorting) and capture a smaller share of those premiums when they do (bargaining). Together, sorting and bargaining account for roughly one-fifth of the cross-sectional gender wage gap, providing the theoretical foundation for the firm-level unit of observation used here.

Beyond these aggregate patterns, individual-level and UK-specific evidence reinforces the structural picture. Bertrand (2018) surveys the glass-ceiling literature, documenting how corporate hierarchies filter women out of senior positions across sectors [20]. Bertrand, Goldin, and Katz (2010) trace the gender earnings gap among young MBAs, showing that hours worked and career interruptions (not initial ability) drive divergence in the first decade after graduation [21]. Cortes and Pan (2018) review how occupational sorting compounds over careers to generate the sectoral pay gap patterns visible in the disclosure data [22]. Andresen and Nix (2022) isolate the child penalty as a causal driver of the career gap using adopting and same-sex couples as a natural experiment [18]. For the UK specifically, Costa Dias, Joyce, and Parodi (2020) attribute most of the UK gender pay gap to children and the differential experience penalties they impose on women’s careers [23], while Joyce and Keiller (2018) document that the gender commuting gap widens sharply in the first decade after childbirth, restricting access to higher-paying jobs [24]. Olsen and Walby (2004) provide a methodological foundation for UK pay gap decomposition that motivates several modelling choices made here [25].

Taken together, these contributions converge on a clear empirical message: **sector, firm, and within-firm pay structure** (especially the female share of top earners) are first-order determinants of the gender pay gap. The features available in the UK disclosure data map directly onto this theoretical apparatus.

## 2.3 Imbalanced Panel Classification

Panel data classification introduces complications absent from cross-sectional ML. First, observa-

tions within a firm are temporally dependent, so standard stratified cross-validation may leak short-range autocorrelation [26]. Second, when the outcome label varies year-to-year within a firm (as `trajectory_label` does here, since it captures a next-year change rather than a firm constant), placing the same firm in both training and validation partitions does not leak a fixed label but may still leak local trend information. Third, class imbalance — Stagnant  $\approx 39\%$ , Improving  $\approx 33\%$ , Worsening  $\approx 28\%$  — can bias classifiers toward the majority class.

Standard remedies include oversampling (SMOTE), undersampling, or cost-sensitive learning via class weights. We adopt class weights rather than resampling to avoid synthetic inflation or information loss. The leakage risk is quantified explicitly by comparing `StratifiedKfold` against `StratifiedGroupKfold` grouped by employer; the macro-F1 gap is at most 0.004 across all models (Section 5), confirming that the panel structure does not materially inflate cross-validation optimism for this label.

## 2.4 Summary and Research Gap

The flagship empirical study of the UK disclosure regime is Blundell, Duchini, Simion, and Turrell [3], in the *American Economic Journal: Economic Policy*. Exploiting the 250-employee threshold as a quasi-experiment in a difference-in-differences design, the authors find that mandatory disclosure closed roughly 19% of the gender pay gap in treated firms, primarily by slowing the wage growth of men rather than accelerating that of women. The effect is concentrated in firms most exposed to public scrutiny: listed firms, consumer-facing sectors, and worse-performing reporters. An earlier version circulated as a CEP Discussion Paper [14]. Comparable evidence from Denmark by Bennedsen, Simintzi, Tsoutsoura, and Wolfenzon (2022) finds that disclosure restrains male wage growth without systematically lifting female wages [19]. Complementary work by Duchini, Simion, and Turrell [5] documents “cracks in the glass ceiling”: an increase in female hiring and promotion into higher quartiles in firms newly subject to the rules. Policy-oriented analysis, including Jones and Kaya [8] and the OECD [10] review of pay-gap reporting in member countries, situates the UK regime within an international comparison, arguing that mandatory, granular, public disclosure is what makes it relatively effective.

What unites these prior contributions is their methodological orientation: they are predominantly **descriptive or causal-econometric**, using OLS, difference-in-differences, and event-study designs to estimate the policy effect of disclosure. They typically pool only a few reporting years and treat the firm-year observation as an outcome rather than as an object to be classified or clustered. To our knowledge, no published study has applied a combined supervised classification and unsupervised clustering pipeline to the full nine-year panel (2017–2025) of UK disclosure data.

The contribution here is therefore threefold. **Methodological breadth:** where prior work asks “did transparency narrow the gap?”, the complementary structural questions are addressed — “what firm features predict the gap?” (RQ1), “who is converging and who is diverging?” (RQ2), and “what latent firm archetypes exist in the data?” (RQ3). **Temporal depth:** nine consecutive reporting years enable trajectory-level modelling that earlier studies, working with shorter panels, could not undertake. **Domain transfer:** applying ML methods that are now standard in computer science but still emerging in labour economics (gradient boosting, PCA,  $K$ -means) to a high-quality administrative panel contributes to the methodological agenda articulated by Mullainathan and Spiess [9] and Strittmatter and Wunsch [12].

## 3 Data Description

### 3.1 Source and Coverage

The raw dataset was obtained directly from the UK Government’s Gender Pay Gap Service,<sup>1</sup> which publishes one CSV file per reporting year starting in 2017. After stacking the annual files covering reporting years 2017–18 through 2025–26, the combined panel consists of roughly 89,000 firm-year observations drawn from 15,429 unique employers, each described by 26 variables capturing pay gap metrics, bonus statistics, pay quartile distributions, and administrative metadata. Each annual file groups into four broad categories: (1) employer identifiers and administrative metadata; (2) hourly pay gap measures; (3) bonus pay gap measures; and (4) pay quartile distributions. Key variables are described in Tables 1–3.

<sup>1</sup><https://gender-pay-gap.service.gov.uk/>

**Table 1:** Variable dictionary: employer identifiers and administrative metadata.

| Variable                  | Type    | Description                                                                | Source          |
|---------------------------|---------|----------------------------------------------------------------------------|-----------------|
| EmployerName              | String  | Name of employer at time of reporting                                      | GPG service     |
| EmployerId                | Integer | Unique identifier; panel unit key                                          | GPG service     |
| CompanyNumber             | Integer | Company registration number (private sector only)                          | Companies House |
| SicCodes                  | String  | UK Standard Industrial Classification codes                                | Companies House |
| PostCode                  | String  | Current registered postcode                                                | GPG service     |
| EmployerSize              | Ordinal | Size band: <250 / 250–499 / 500–999 / 1,000–4,999 / 5,000–19,999 / 20,000+ | Self-reported   |
| SubmittedAfterTheDeadline | Boolean | TRUE if submitted after statutory deadline                                 | GPG service     |
| DateSubmitted             | Date    | Actual submission date                                                     | GPG service     |

**Table 2:** Variable dictionary: hourly and bonus pay gap measures.

| Variable                | Description                                     |
|-------------------------|-------------------------------------------------|
| DiffMeanHourlyPercent   | Mean % hourly pay gap; positive = men paid more |
| DiffMedianHourlyPercent | Median % hourly pay gap; robust to outliers     |
| DiffMeanBonusPercent    | Mean % bonus pay gap                            |
| DiffMedianBonusPercent  | Median % bonus pay gap                          |
| MaleBonusPercent        | % of men receiving a bonus                      |
| FemaleBonusPercent      | % of women receiving a bonus                    |

**Table 3:** Variable dictionary: pay quartile distributions (all self-reported).

| Variable                  | Description                             |
|---------------------------|-----------------------------------------|
| MaleLowerQuartile         | % of men in lowest pay quartile         |
| FemaleLowerQuartile       | % of women in lowest pay quartile       |
| MaleLowerMiddleQuartile   | % of men in lower-middle pay quartile   |
| FemaleLowerMiddleQuartile | % of women in lower-middle pay quartile |
| MaleUpperMiddleQuartile   | % of men in upper-middle pay quartile   |
| FemaleUpperMiddleQuartile | % of women in upper-middle pay quartile |
| MaleTopQuartile           | % of men in top pay quartile            |
| FemaleTopQuartile         | % of women in top pay quartile          |

## 3.2 Data Preprocessing

### 3.2.1 Removal of Non-Analytical Variables

Variables consisting exclusively of administrative details or logistical metadata — `CompanyLinkToGPGInfo`, `CurrentName`, `SubmittedAfterTheDeadline`, `DueDate`, `DateSubmitted`, `ResponsiblePerson`, and `Address`

— do not support statistical inference for pay gap trends, increase dimensionality unnecessarily, and provide no predictive value. They are dropped prior to modelling.

### 3.2.2 SIC Code Cleaning

The `SicCodes` field often contains multiple codes separated by commas, line breaks, or extraneous text. We standardised the industrial classification by extracting a single representative numeric code from each entry using pattern matching and selecting the highest value. The resulting code is then mapped to the 21 UK SIC divisions, yielding the categorical feature `Nature_of_business` suitable for one-hot encoding.

### 3.2.3 Employer Size Encoding

The `EmployerSize` variable is provided as categorical text strings (e.g., “500 to 999”). We converted these into a numerical ordinal scale, mapping each band to an integer from 0 (fewer than 250 employees) to 5 (20,000 or more), with “Not Provided” stored as null. This transformation ensures models recognise the hierarchical relationship between size bands.

### 3.2.4 Postcode Geocoding

The primary dataset lacked structured geographic information beyond raw postcode strings. We imported a secondary reference dataset mapping UK outcodes to town, country, and region, then matched each employer’s outcode prefix (the portion before the space) to extract and merge `town`, `country`, and `region` into the main dataset.

### 3.2.5 Outlier Detection and Winsorisation

The dataset contained significant outliers and erroneous values, including impossible percentages and extreme spikes (e.g., bonus gaps reaching millions of percent, and placeholder values such as  $-999,999,999$ ). These anomalies would disproportionately bias linear models and distort distance-based calculations. We implemented winsorisation after identifying the extent of the problem via inter-quartile range (IQR) analysis, capping the four core pay and bonus variables at the 5th and 95th percentiles. This preserves the overall sample size and maintains the signal of high-inequality companies without allowing extreme noise to destabilise model coefficients.

### 3.2.6 Trajectory Label Construction

Single-year snapshots do not reveal whether a company’s gender pay gap is trending upward or downward. We constructed `avg_annual_change`, the **next-year change** in `DiffMedianHourlyPercent` for each employer-year pair:  $\text{gap}_{t+1} - \text{gap}_t$  within each employer. Based on this change, we assigned `trajectory_label`: **Improving** ( $\leq -1.5$  pp), **Worsening** ( $\geq +1.5$  pp), **Stagnant** otherwise, and **Insufficient Data** when the next observation is unavailable (the employer’s final reporting year). The label is a *per-(employer, year)* quantity that varies within a firm from year to year; it is not a fixed firm-level attribute.

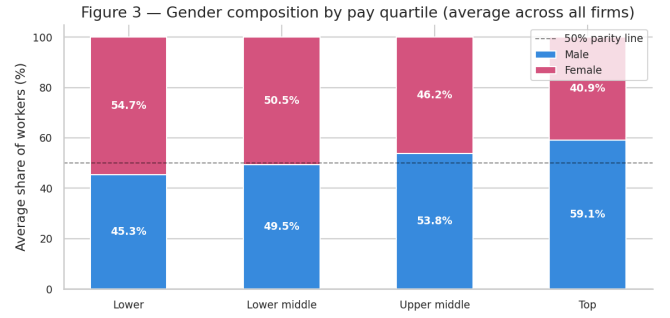
### 3.2.7 Longitudinal Cohort

In addition to the full panel (`dataset_total.csv`), we isolated a longitudinal cohort by retaining only companies with at least three consecutive years of reporting (approximately 11,780 employers, roughly 83,000 rows). The supervised classification models use the full panel; the cohort supports exploratory trajectory analyses.

## 3.3 Exploratory Data Analysis

### 3.3.1 Univariate Analysis

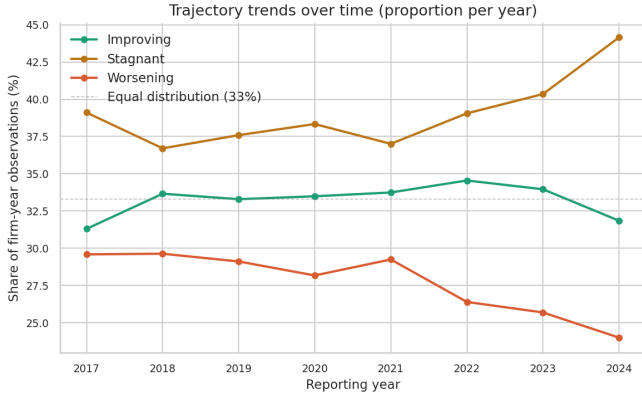
The univariate analysis focuses on the features most directly relevant to the research question including the quartile composition, one of the core predictive features; and the trajectory label, the classification target.



**Figure 1:** Gender composition by pay quartile (average across all firms)

The quartile composition of firms reveals the structural pattern most directly relevant to this analysis. As shown in Figure 1, women occupy 54.7% of lower-paid positions but only 40.9% of top-quartile roles, with male overrepresentation increasing progressively up the pay distribution. This gradient confirms that the hourly pay gap is not only driven by unequal pay for equal work, but also by the systematic concentration of women in lower-paying positions which then lead to lower female hourly pay averages; a distinction that was also noted by [2]. Employer size and sector composition further contextualise the sample: the majority of reporting firms fall in the 250–499 employee band (43.5%), meaning findings are disproportionately shaped by small-to-medium enterprises, while the top five industries by observation count — administration, education, manufacturing, wholesale & retail, and professional services — span a wide range of known gender gap profiles, lending the dataset analytical breadth.

Figure 2 reveals a noticeable shift in trajectory over the reporting period. Pooled across the 2023–2024 reporting years, the stagnant share has risen to roughly 42%, up from approximately 38% in 2017, while the share of worsening firms has declined from around 30% to 25%. The proportion of improving firms has remained relatively flat throughout the period at around 33%. This shows that although fewer firms are actively worsening, transparency alone does not appear to be a strong enough driver of change. The growing stagnant category suggests that mandatory reporting has led firms to disclose the state of their organisation without necessarily translating that disclosure into meaningful action.



**Figure 2:** Trajectory trends over time (proportion per year)

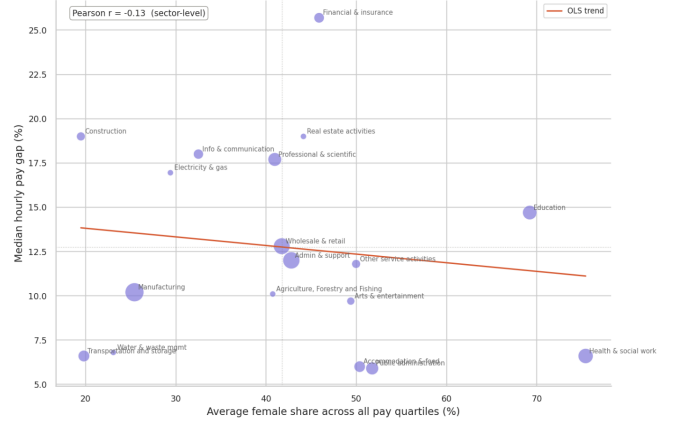
### 3.3.2 Bivariate Analysis

The bivariate analysis focuses on two structural dimensions central to understanding gender pay inequality: sector and female representation.

The most immediate source of cross-firm variation is sector. Finance and Insurance and Professional and Scientific Services record median gaps above 20 pp, while Health and Social Work sits near 7 pp (Figure 3). Beyond differences in central tendency, distributional shape is informative: Finance exhibits a long right tail with firms exceeding 40 pp, while Health and Social Work shows a tightly concentrated distribution consistent with collectively-bargained pay structures.

Figure 4 plots sector-level average female share across all pay quartiles against the median pay gap, with bubble size reflecting the number of firms per sector. The negative correlation ( $r = -0.13$ ) is modest but consistent: sectors with lower female representation tend to record higher gaps. Finance and Insurance stands out: despite a near-average feminisation rate of around 45%, it records the highest median gap at over 25 pp, suggesting that in this sector the gap is driven more by within-level pay differences than by underrepresentation alone. Health and Social Work combines high female representation (above 75%) with one of the lowest gaps ( $\approx 6.5$  pp). Education presents an interesting anomaly: the highest feminisation rate of all sectors ( $\approx 70\%$ ) yet a gap of nearly 15 pp, pointing to persistent vertical segregation within an otherwise female-dominated sector. Overall, the weak sector-level correlation suggests that feminisation rate alone is insufficient to explain pay gap variation and that firm-level

structural factors play an equally important role.



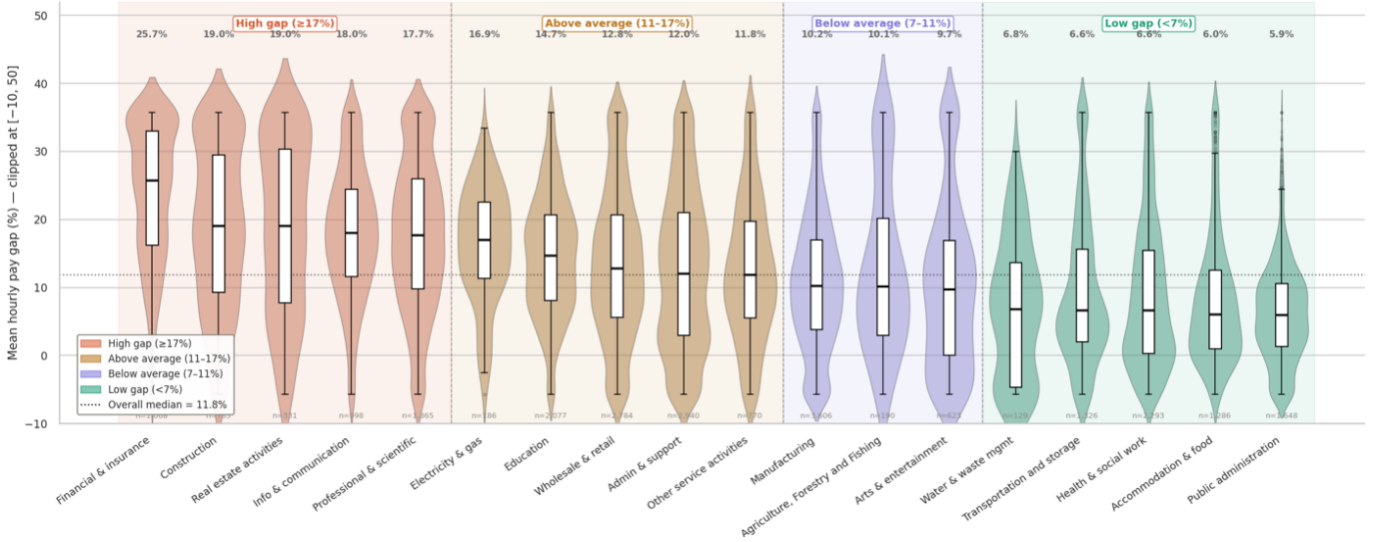
**Figure 4:** Sector-level average female share (all pay quartiles) vs. median hourly pay gap. Bubble size reflects number of firms.  $r = -0.13$ .

## 4 Methodology

### 4.1 Problem Framing

The supervised learning task is a multiclass classification problem. The goal is to predict whether a firm’s gender pay gap will improve, stagnate, or worsen in the following reporting year, using only information available in the current year. The target variable takes one of three values (*Improving*, *Stagnant*, or *Worsening*), defined as the sign of the change in a firm’s median hourly pay gap between the current and following reporting year (threshold at  $\pm 1.5$  percentage points). This variable varies from year to year within the same firm, and the threshold was chosen to separate meaningful directional movement from the kind of small fluctuations that are likely due to reporting noise rather than genuine organisational change. Firm-year rows for which the next-year change is undefined (the employer’s final reporting year) carry an *Insufficient Data* label and are dropped from training and evaluation, reducing the working sample to roughly 74,000 firm-year observations.

The prediction task was designed carefully to avoid data leakage. All features used to predict a firm’s trajectory in year  $t + 1$  come from year  $t$  observations only, so no future information enters the training process. This makes the task a genuine forecasting exercise: predicting what a firm will do next year based on what we can observe today, which is directly relevant to the policy goal of identifying firms at risk of persistent inequality before it becomes entrenched.



**Figure 3:** Distribution of hourly pay gap by sector (violin plots), ordered by median gap.

The dataset is slightly imbalanced, with the Stagnant category making up the larger share of observations in both training and test sets at approximately 40%, followed by Improving at 33% and Worsening at 28%. Rather than discarding real observations through downsampling or generating synthetic ones through upsampling, class weighting was applied across all models. This tells each model to penalise misclassification of minority classes more heavily during training, correcting the imbalance without altering the underlying data.

**Table 4:** Class distribution in training and test sets after preprocessing

| Class     | Train  |      | Test  |      |
|-----------|--------|------|-------|------|
|           | N      | %    | N     | %    |
| Improving | 19,063 | 33.3 | 5,597 | 33.2 |
| Stagnant  | 21,714 | 38.0 | 7,046 | 41.8 |
| Worsening | 16,380 | 28.7 | 4,229 | 25.0 |

## 4.2 Data Preparation for Modelling

Three variables required imputation before modelling. Missing values in `DiffMedianBonusPercent` (affecting 679 firms) were confirmed to be random reporting omissions rather than structural absences, as they were evenly spread across years, industries, and employer sizes, and occurred exclusively among firms that reported paying bonuses. These were imputed using the training set median, and the same training median was applied to the test set, ensuring that no information from the test period

influenced the imputation. For firms with no bonus schemes (`PaysBonuses` = 0), both `bonus_gap` and `PaysBonuses` were set to zero.

The missing quartile data, affecting 370 firms and concentrated entirely in 2020 and 2021, was attributed to the UK government’s temporary suspension of mandatory gender pay gap reporting during the COVID-19 pandemic [27]. To address these missing values, forward-fill imputation was applied within each firm’s time series, using the firm’s own prior-year quartile composition as the best available estimate. This is more appropriate than dataset-wide median imputation because workforce composition tends to be stable year-to-year within a given firm. Any firms whose first report fell in 2020 or 2021, and so had no prior data to fill from, were dropped entirely, affecting fewer than 0.5% of observations. Missing values in the encoded employer size feature were addressed by assigning the “Not Provided” label a dedicated numeric code rather than imputing.

Industry sector and geographic region were one-hot encoded, with the first category dropped to avoid perfect multicollinearity. Numerical features were standardised using `StandardScaler`, which was necessary for logistic regression, which uses gradient-based optimisation that is sensitive to differences in feature scale; otherwise larger numerical ranges would dominate the optimisation and the algorithm would fail to converge. The scaler was fitted on the training set only and applied to the test set using the training parameters, keeping the scaling process free from leakage.

### 4.3 Train/Test Split

A time-based train/test split was implemented rather than random sampling. Models were trained on data from 2017 to 2022 and tested on 2023 and 2024, giving an approximate 77/23 split. Random splitting was avoided because it would allow the model to train on 2023 data and test on 2019 data, which breaks the temporal logic of forecasting and would produce misleadingly optimistic performance estimates [26]. The time-based approach genuinely mirrors real deployment: a model trained on historical reports is used to predict the next year’s outcomes.

The 2025 reporting year was excluded from both training and evaluation because reporting for that period was still ongoing at the time of data extraction, making the sample incomplete. As reporting completes, 2025 represents a natural out-of-sample validation set for future work.

Because the same employer can appear across multiple reporting years, the majority of training rows belong to firms that are also present in the test set. This overlap is a potential source of leakage (autocorrelation in firms’ consecutive-year pay gaps). To quantify this, a `StratifiedGroupKFold` cross-validation blocked on employer identity was run alongside a standard `StratifiedKFold` on the training set, with results reported in Section 5.

Macro-averaged F1 score was chosen as the primary metric over accuracy and weighted F1 because accuracy is misleading under class imbalance, and weighted F1 would disproportionately reward performance on the majority class. Macro F1 averages performance equally across all three classes, ensuring that the model’s ability to identify improving and worsening firms counts as much as its ability to identify stagnant ones.

### 4.4 Model Selection and Rationale

Five models were trained in order of increasing complexity, alongside a stratified dummy classifier as a chance baseline (predicting classes in proportion to their training set frequency without using any features) which establishes the minimum performance threshold any learned model must meaningfully exceed. Before holdout evaluation, hyperparameters for Random Forest, Gradient Boosting, and Logistic Regression were tuned using `RandomizedSearchCV` with inner cross-validation via `StratifiedGroupKFold` (5

folds, grouped by employer), scored on macro-F1. To keep computation manageable, the search ran on a stratified 20,000-row subsample of the training set, more than sufficient to rank hyperparameters. The winning parameters were then applied to models refitted on the full training set.

Logistic Regression serves as the baseline because it is simple, interpretable, and a necessary reference point before adding complexity. Two regularised variants follow: Ridge (L2), which shrinks coefficients smoothly and stabilises the fit under correlated predictors, and Lasso (L1), which drives the weakest coefficients to exactly zero and performs implicit feature selection across the 209-column feature space. Random Forest was chosen as the first tree-based model because it captures non-linear relationships and feature interactions automatically, and produces feature importance scores that directly address the research question. Gradient Boosting extends this by building trees sequentially, each correcting the errors of the last, making it better suited to datasets where patterns are subtle. Support Vector Machines were rejected for lacking feature importance scores, and neural networks for offering no interpretability advantage over tree-based models on tabular data of this scale. Feature importance is assessed through three complementary lenses: permutation importance on the best tree (model-agnostic), standardised linear coefficients (sign and magnitude), and SHAP values on Gradient Boosting (per-row contribution, capturing interactions), with findings reported in the Supervised Learning subsection under Results Section.

## 5 Results

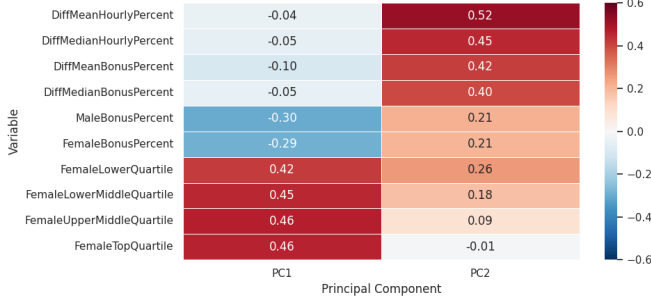
### 5.1 Unsupervised Learning

To uncover latent patterns in the data without using predefined sector categories (Nature of business), we applied unsupervised learning techniques. Specifically, PCA for dimensionality reduction and K-Means clustering were performed to determine whether salary and bonus gap properties intrinsically group companies into distinct behavioral profiles.

#### 5.1.1 Principal Components Analysis (PCA)

Prior to clustering, Principal Component Analysis (PCA) was applied to the ten standardised numerical variables capturing pay gap magnitude

and gender representation across salary quartiles. As demonstrated by the scree plot, the first two principal components account for 65.2% of the total variance (PC1: 39.8%, PC2: 25.4%). With just four components capturing 91.4% of the information, it is clear that the ten original features are highly correlated and justify the reduction of the feature into a lower-dimensional space with minimal information loss. The loadings heatmap (figure 5) reveals a clear and logical division of the data.



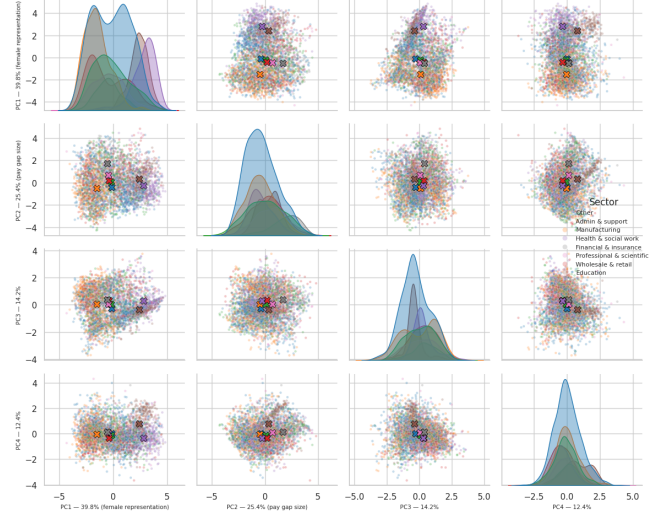
**Figure 5:** PCA loadings, all retained components (red = positive, blue = negative contribution).

In PCA, loadings indicate how strongly each original variable influences a component, with values near  $\pm 0.5$  showing a strong impact and values near zero showing almost none. This metric allows us to clearly interpret the first two components:

- PC1 acts as a measure of Female Representation. It is strongly driven by the four female quartile variables (loadings between 0.42 and 0.46) and ignores the pay gap variables. A firm scoring high on PC1 successfully employs women across all hierarchical and pay levels.
- PC2 acts as a measure of the Pay and Bonus Gap. Conversely, it is driven entirely by the four gap variables (loadings between 0.40 and 0.52) and ignores the quartile variables. It captures the pure magnitude of the financial gap, completely independent of workforce representation.

This clean separation is substantively important: it confirms empirically that representation and pay gap are two distinct dimensions of gender inequality. A firm can employ many women while still paying them substantially less: the two axes capture structurally different problems requiring different policy responses. The 4D pair plot (figure 6) further illustrates this, showing that sector centroids are meaningfully spread across the PC1–PC2 plane, with

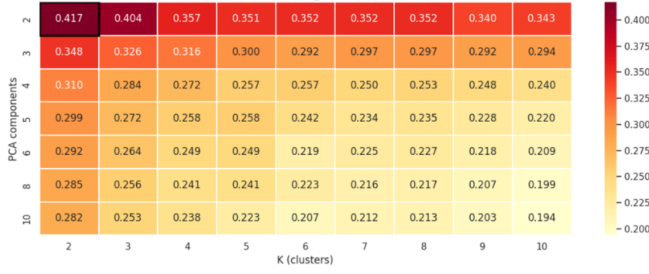
Health & Social Work and Education clustering toward high PC1 scores, while Financial & Insurance and Manufacturing occupy lower PC1 positions with higher PC2 scores.



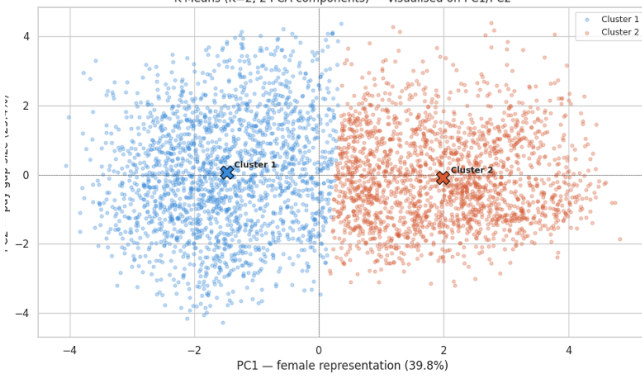
**Figure 6:** Companies projected in 4D PCA space (x markers indicate sector centroids).

### 5.1.2 K-means Clustering

To partition the observations into distinct groups, we applied K-Means clustering. This clustering was performed on the PCA-reduced space rather than the original dataset, allowing the model to operate on the most informative dimensions while filtering out underlying noise. To objectively determine the optimal model architecture, we conducted a joint grid search evaluating all combinations of PCA components (2 to 10) and cluster counts (k from 2 to 10). The models were evaluated using the silhouette score, a metric ranging from -1 (poor separation) to 1 (perfect separation), that quantifies how similar an observation is to its own cluster compared to neighboring ones. As shown in the full grid (figure 7), silhouette scores decline monotonically for any  $k > 2$ , indicating that finer partitions do not represent genuinely separable structures in the data. Consequently, the optimal configuration was identified as  $k=2$  clusters using 2 PCA components, which yielded a maximum silhouette score of 0.417.



**Figure 7:** Silhouette score grid across PCA components and cluster counts.



**Figure 8:** K-means clusters visualised on PC1/PC2.

The resulting clusters, visualized in Figure 8 along the PC1–PC2 plane, reveal a clear horizontal separation driven primarily by PC1 (female representation). Note that this clustering partition was performed on a refined analytical sample of approximately 79,000 observations, reflecting the full initial panel of roughly 89,000 rows excluding the incomplete 2025 reporting year.

Within this sample, Cluster 1 ( $n \approx 39,500$ ) groups firms with high female representation and a relatively smaller pay gap. Its sector composition is dominated by Health & Social Work, Education, and Admin & Support, sectors historically characterised by female-majority workforces and more compressed pay distributions. Cluster 2 ( $n \approx 39,500$ ) concentrates firms with lower female representation and a larger pay gap, drawing predominantly from Manufacturing, Wholesale & Retail, and Other services, with a notably higher share of Financial & Insurance and Professional & Scientific firms relative to Cluster 1.

This two-cluster solution carries a strong substantive meaning: gender inequality across UK firms is characterized by a stark structural divide. Essentially, a firm’s profile is binary: it either embeds women at all pay levels with controlled wage gaps, or it exhibits

a ‘glass ceiling’ effect, excluding women from top roles while maintaining severe disparities. Because the pay gap magnitude (PC2) does not drive this separation, it suggests that the size of the gap is a continuous variable rather than a defining structural feature for classification. To explore the finer nuances within the second cluster (e.g., distinguishing between firms with high bonus gaps versus strict glass ceilings), we transition to the regression and classification techniques detailed in the next section.

## 5.2 Supervised Learning

### 5.2.1 Cross-Validation Leakage Diagnostic

As outlined in the methodology, because employers report for multiple consecutive years, a standard stratified fold may place the same firm in both training and validation, leaking short-range auto-correlation in the gap. Every model was therefore evaluated under two cross-validation schemes on the training set (2017–2022): plain **StratifiedKfold** (SKF) and **StratifiedGroupKfold** grouped by employer (GKF). The gap in macro-F1 between the two schemes quantifies how much the model exploits firm identity rather than structural features.

**Table 5:** Cross-validation leakage diagnostic. SKF = StratifiedKfold; GKF = StratifiedGroupKfold by employer. Gap = SKF minus GKF macro-F1.

| Model               | SKF   | GKF   | Gap   |
|---------------------|-------|-------|-------|
| Dummy (stratified)  | 0.331 | 0.331 | 0.000 |
| Logistic Regression | 0.428 | 0.425 | 0.003 |
| Ridge LogReg (L2)   | 0.428 | 0.425 | 0.003 |
| Lasso LogReg (L1)   | 0.434 | 0.430 | 0.004 |
| Random Forest       | 0.468 | 0.464 | 0.004 |
| Gradient Boosting   | 0.484 | 0.483 | 0.001 |

Gaps range from 0.000 to 0.004 across all models, effectively zero. The result is expected: **trajectory\_label** is a per-*(employer, year)* next-year change, not a firm-level constant, so placing a firm in both training and validation partitions does not allow the model to memorise a fixed label. The model relies on sector, size, quartile, and bonus structure rather than firm identity, and would therefore generalise to previously unseen firms. The grouped and ungrouped scores are both honest reads of generalisation ability.

### 5.2.2 Holdout Performance (2023–2024)

All models were refit on the full training set (2017–2022, approximately 57,000 observations) and scored on the held-out test set (2023–2024, 16,869 observations). Table 6 reports per-class F1, balanced accuracy, and a bootstrap 95% confidence interval on macro-F1 (1,000 resamples).

The stratified dummy scores  $\text{macro-F1} = 0.329$ ; always predicting *Stagnant* would score approximately 0.19. The meaningful range for a three-class problem with class shares of approximately 0.33, 0.39, and 0.28 is 0.33 (no skill) to 1.0 (perfect). Gradient Boosting reaches **0.502** [0.494, 0.510], clearly above the floor but not high in absolute terms. Random Forest is second at 0.479 [0.470, 0.486]. The logistic family clusters between 0.435 and 0.438, reflecting the non-linear high-dimensional boundary that a log-linear model cannot recover.

Worsening is the hardest class across all models: Gradient Boosting achieves  $\text{F1} = 0.391$  for Worsening versus 0.572 for Stagnant and 0.544 for Improving, a gap that runs consistently across both the logistic and tree families. The holdout macro-F1 (0.502) sits slightly above the grouped-CV figure (0.483), reflecting the 2023–2024 test years having a larger Stagnant share than the training years rather than any firm-overlap optimism.

### 5.2.3 Confusion Structure

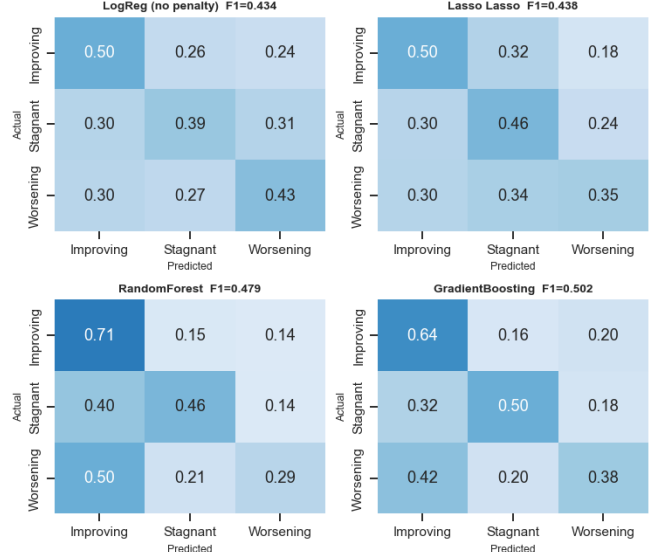
Figure 9 shows row-normalised confusion matrices for all five non-dummy models. For Gradient Boosting specifically, 8,148 of the 16,869 test observations are misclassified (48.3%). Table 7 decomposes the errors.

**Table 7:** Absolute confusion matrix, Gradient Boosting on the 2023–2024 holdout. Row = true class; column = predicted class.

| True \ Predicted  | Imp.         | Stag.        | Wors.        |
|-------------------|--------------|--------------|--------------|
| Improving (5,595) | <b>3,594</b> | 906          | 1,095        |
| Stagnant (7,045)  | 2,252        | <b>3,525</b> | 1,268        |
| Worsening (4,229) | 1,776        | 851          | <b>1,602</b> |

Two patterns dominate. First, Stagnant acts as a gravitational attractor: the single largest off-diagonal cell is true Stagnant predicted as Improving (2,252 cases), reflecting the model’s tendency to over-call improvement for firms near the  $\pm 1.5$  pp

boundary. Second, direction flips are common: Improving predicted as Worsening (1,095) together with Worsening predicted as Improving (1,776) account for **2,871 errors, 35.2% of all misclassified rows**. The  $\pm 1.5$  pp threshold discretises a continuous next-year change; firms near the cut-off are inherently ambiguous and inflate off-diagonal cells regardless of model quality.



**Figure 9:** Row-normalised confusion matrices on the 2023–2024 holdout. Diagonal = correctly classified share. Off-diagonal = share of true class predicted into each other class.

### 5.2.4 Per-Segment Error Rates

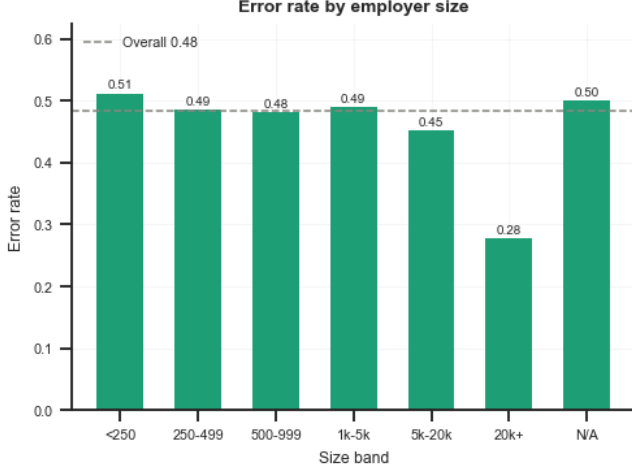
Error rate ( $1 - \text{accuracy}$ ) and macro-F1 are reported by year, employer size, and sector for Gradient Boosting predictions on the 2023–2024 holdout.

**Year.** Error rate is 0.480 in 2023 and 0.489 in 2024 (macro-F1: 0.506 and 0.495, respectively). The near-identical figures indicate no concept drift between the two test years: the structural drivers of where a firm sits in the pay-gap distribution did not shift, and the model generalises equally to both.

**Employer size.** Error rate falls sharply with firm size. The smallest category (fewer than 250 employees) reaches an error rate of **0.512**, while the largest (20,000 or more) reaches **0.278**. For small firms, a small number of hires or departures can swing the year-to-year gap by several percentage points, making the trajectory label mechanically volatile and hard to recover from a single cross-section. For large firms, workforce composition moves slowly and the gap

**Table 6:** Holdout performance on the 2023–2024 test set (16,869 observations). Bootstrap 95% CI on macro-F1 from 1,000 resamples. Best model in bold.

| Model                    | Macro-F1     | 95% CI                | F1-Imp.      | F1-Stag.     | F1-Wors.     | Accuracy     | Bal. Acc.    |
|--------------------------|--------------|-----------------------|--------------|--------------|--------------|--------------|--------------|
| Dummy (stratified)       | 0.329        | [0.323, 0.336]        | 0.331        | 0.392        | 0.266        | 0.337        | 0.330        |
| Logistic Regression      | 0.435        | [0.427, 0.442]        | 0.478        | 0.443        | 0.383        | 0.438        | 0.442        |
| Ridge LogReg (L2)        | 0.435        | [0.427, 0.443]        | 0.478        | 0.443        | 0.383        | 0.438        | 0.442        |
| Lasso LogReg (L1)        | 0.438        | [0.430, 0.445]        | 0.476        | 0.481        | 0.355        | 0.448        | 0.439        |
| Random Forest            | 0.479        | [0.470, 0.486]        | 0.550        | 0.541        | 0.344        | 0.503        | 0.489        |
| <b>Gradient Boosting</b> | <b>0.502</b> | <b>[0.494, 0.510]</b> | <b>0.544</b> | <b>0.572</b> | <b>0.391</b> | <b>0.517</b> | <b>0.507</b> |



**Figure 10:** Gradient Boosting error rate by year (top), employer size (middle), and sector (bottom). Macro-F1 reported alongside each segment.

trajectory is more predictable.

**Sector.** The hardest sectors are Agriculture, Forestry and Fishing (error rate 0.560), Information and Communication (0.541), and Construction (0.535). The easiest are Accommodation and Food Service Activities (0.361). The pattern reflects two distinct mechanisms: sectors with compressed, grade-based pay structures cluster tightly around Stagnant near zero, giving the model an easy baseline; bonus-heavy and project-based sectors combine high absolute gaps with idiosyncratic firm-level pay decisions that the feature set cannot observe, producing both higher errors and higher within-sector variance.

### 5.3 Feature Importance (RQ3)

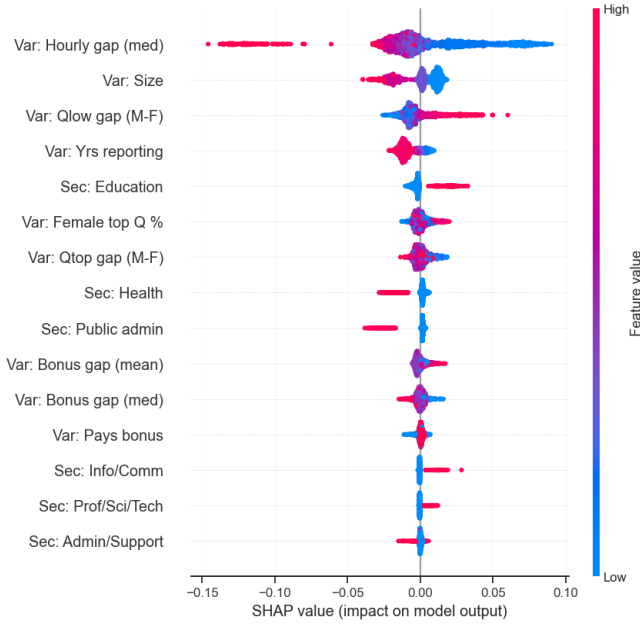
This subsection addresses RQ3: which firm characteristics most strongly predict the *direction* of next-year change in the median hourly pay gap. Three complementary lenses are used to identify which features drive the trajectory classifier’s predictions

(Figure 11).

**Permutation importance (Gradient Boosting).** Permutation importance measures the drop in macro-F1 when a feature’s values are randomly shuffled. The top contributors are the glass-ceiling quartile features (`quartile_gap_top`, `FemaleTopQuartile`), bonus measures (`bonus_gap`, `DiffMedianBonusPercent`), the current gap level (`DiffMedianHourlyPercent`), and sector dummies for Finance and Professional Services. Region dummies and `EmployerSize_Encoded` alone rank low, consistent with size being a moderator of sector effects rather than an independent predictor.

**Logistic coefficients.** The log-odds difference (Worsening minus Improving) on standardised features captures the direction of each feature’s linear push. Finance, Construction, and Professional Services dummies carry positive coefficients, pushing toward Worsening; Health, Education, and Public Administration carry negative coefficients, pushing toward Improving or Stagnant. Among continuous features, a higher `bonus_gap` and a higher current `DiffMedianHourlyPercent` both increase the log-odds of Worsening relative to Improving.

**SHAP values (Random Forest, Worsening class).** Because sklearn’s TreeExplainer does not support multiclass Gradient Boosting, SHAP is computed on Random Forest. Figure 11 shows the per-row SHAP beeswarm for the *Worsening* class. The top contributor is `DiffMedianHourlyPercent`: high current-gap rows (red) sit on the left, pushing log-odds away from Worsening, consistent with mean reversion in firms already at the high end of the distribution. `Size`, `quartile_gap_lower`, and `FemaleTopQuartile` rank immediately below, confirming that firm scale and vertical segregation shape the trajectory signal. The corresponding pooled mean-|SHAP| ranking across all three classes



**Figure 11:** SHAP beeswarm for the *Worsening* class (Random Forest,  $n = 2,000$  test samples). Each dot is one firm-year; horizontal position = SHAP value (impact on log-odds of Worsening); colour = feature value (red high, blue low).

(computed in the notebook but not reproduced here) places the same features at the top, indicating the Worsening-class drivers are also the dominant overall predictors.

## 6 Conclusion

**Addressing RQ1 (Latent structural archetypes)**, our unsupervised learning pipeline reveals that UK firms generally bifurcate into two distinct structural profiles: one that successfully integrates women across all hierarchical levels with compressed wage gaps, and a second "glass ceiling" archetype that combines severe pay disparities with the exclusion of women from top-quartile roles. Crucially, the PCA confirms that female representation and the financial pay gap operate as two distinct dimensions of inequality.

**Regarding RQ2 (Trajectory prediction)**, Gradient Boosting achieves the highest holdout macro-F1 at 0.502, against a stratified dummy floor of 0.329. Tree ensembles outperform linear models because the decision boundary separating the three trajectory classes is non-linear in the quartile, bonus, and sector feature space: a linear log-odds surface cannot capture the interaction between, for example, a high

bonus gap and a Finance sector dummy. Random Forest is a close second; the logistic family loses roughly 0.07 macro-F1 points to Gradient Boosting because it cannot recover these non-linearities.

However, the modest absolute performance is not primarily a modelling failure: it is the central finding. A firm's contemporaneous structure carries strong information about *where* its gender pay gap sits today but weak information about *which direction it will move next year*. Mandatory disclosure has produced only marginal, unsystematic year-to-year movement across the reporting population, leaving little forecastable trend signal for any model to recover. The gap between the best model and the dummy floor (approximately 0.17 macro-F1 points) is real but narrow, and confirms that next-year gap direction is a hard prediction problem by construction.

**Finally, in response to RQ3 (Structural drivers)**, the structural features that do emerge as predictive are consistent with established explanations of the gender pay gap. Vertical segregation (proxied by `quartile_gap_top` and `FemaleTopQuartile`) is the dominant non-sector signal: firms whose senior pay quartile is already badly skewed toward men are more likely to worsen, consistent with the self-reinforcing nature of glass-ceiling effects documented in Blau and Kahn [2] and the IFS Deaton Review [13]. Bonus culture (`bonus_gap`, `DiffMedianBonusPercent`) ranks alongside the quartile features: in sectors where bonuses form a large share of total pay, a widening bonus gap is an early signal of a widening hourly gap. Region alone adds no independent signal beyond sector composition. Compliance maturity (`years_reporting`) is negligible, suggesting that firms that have reported longer are not on systematically improving paths.

### 6.1 Limitations

**Feature-target mismatch.** The features describe a firm's contemporaneous structure in a given year. This structure is strongly informative about the *level* of the gender pay gap but only loosely informative about *which direction it moves next year*. The model's modest accuracy reflects a genuine scarcity of forecastable structure rather than primarily a modelling failure. Adding richer temporal features (multi-year gap slope, lagged quartile changes) or

within-firm occupational composition data could raise the ceiling, but the fundamental volatility of year-to-year direction changes limits any snapshot-based model.

**Label discretisation.** The `trajectory_label` is constructed by thresholding a continuous next-year change at  $\pm 1.5$  percentage points. The threshold is empirically motivated but arbitrary. Firms near the boundary are inherently ambiguous regardless of model quality: a true next-year change of +1.4 pp and one of +1.6 pp receive different labels but are practically indistinguishable. Direction-flip errors (35.2% of all misclassified rows) partly reflect this boundary artefact rather than model error in any meaningful sense.

**No within-firm occupational data.** The mandatory disclosure dataset is a firm-level aggregate. It cannot separate a like-for-like pay gap from a composition gap driven by occupational segregation, nor a deliberate pay correction from a workforce-mix shift. A firm that reduces its gender pay gap by hiring more women into low-paid roles and a firm that achieves genuine pay equity both receive an *Improving* label. This conflation limits both the interpretability of predictions and the policy conclusions that can be drawn from feature importance.

**Data sparsity and firm overlap.** Reporting for the 2024–2025 cycle is still open, so the test set is effectively limited to 2023–2024. Approximately 87% of training firms also appear in the test set; however, the grouped cross-validation diagnostic shows a leakage gap of at most 0.004 macro-F1, confirming that firm overlap leaks only via short-range gap autocorrelation and not via firm-identity memorisation.

## 6.2 Discussion

**Policy targeting by firm size.** The per-segment analysis reveals a stark size gradient: error rate falls from 0.512 for employers with fewer than 250 employees to 0.278 for those with 20,000 or more. A regulator applying this model should not treat small-firm predictions as actionable, since trajectories at that scale are mechanically volatile and driven by individual hiring decisions rather than structural pay policy. The model is most reliable for large, well-categorised firms, where workforce composition moves slowly and the structural features

in the disclosure dataset are genuinely informative. This asymmetry also bears on disclosure policy: the 250-employee reporting threshold means the dataset under-represents precisely the segment where prediction is hardest, making systematic monitoring of small-employer pay gaps infeasible from mandatory disclosure data alone.

**Region as a sector proxy.** Region dummies carry negligible independent predictive weight once sector and firm-level pay structure are controlled for. This aligns with the EDA finding that regional variation in the gender pay gap is largely a function of sectoral composition: the high gap observed in London reflects the concentration of Finance and Professional Services firms, not a distinctly regional pay dynamic. This result implies that regional policy interventions are unlikely to be well-targeted unless they specifically address the sector-level mechanisms (bonus culture, vertical segregation) driving the gap in that geography.

**Compliance maturity and disclosure effectiveness.** The feature `years_reporting` counts how many years a firm has been subject to mandatory disclosure. Its near-zero permutation importance is a substantively important null result: firms that have been reporting for longer are not on systematically improving trajectories relative to newer reporters. If reputational pressure and public accountability were functioning as intended, longer-reporting firms should show more consistent improvement. The absence of this pattern, combined with the model’s overall low macro-F1, is consistent with the view that mandatory disclosure alone is insufficient to produce systematic, sustained reduction in the gender pay gap, and that structural interventions targeting occupational segregation and bonus design are necessary complements.

## 6.3 Outlook and Future Work

Several directions could extend the analysis presented here.

**Regression on gap level.** The classification framing discards the continuous variation in next-year gap movement. A regression target predicting the magnitude of the gap rather than its direction would recover this information and complement the trajectory analysis.

**Multi-year trajectory modelling.**

`trajectory_label` captures a single next-year change. A multi-year slope over three or more consecutive years would be less sensitive to noise at the  $\pm 1.5$  pp boundary and better aligned with the policy question of sustained improvement versus a one-year fluctuation.

#### **Firm-level fixed effects and sequential models.**

The current models treat each firm-year observation independently. Recurrent neural networks or fixed-effect panel models could exploit the sequential nature of the data and capture how last year’s gap level influences this year’s direction of change.

**Occupational composition data.** The mandatory disclosure dataset is a firm-level aggregate and cannot separate a like-for-like pay gap from a composition-driven gap. Linking disclosure records to matched employer–employee data would allow within-occupation gender pay comparisons and sharpen the interpretation of structural predictors.

**Full 2024–2025 reporting year.** Reporting for the 2024–2025 cycle was still open at the time of data extraction; the test set is therefore effectively limited to 2023–2024. As the complete data becomes available, the holdout set can be extended by one year, providing a stricter out-of-time evaluation.

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